From Rhetorical Labelling to Targeted Verification

False alarms, abstention, and negation checking in a language-model classifier

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Abstract

Automatic identification of rhetorical devices can support critical reading, but it can also produce unwarranted attributions when a system confuses charged vocabulary with the actual use of a device. This report examines a classifier for six rhetorical categories, evaluated on 60 contrastive pairs totalling 120 texts. The initial classifier identified the technique correctly in 55 of the 60 positive examples, but produced 39 false alarms on the 60 controls. A version with instructions about negation and quotation, together with the option to abstain, reduced the false alarms to 16, while correct positive classifications fell to 51. A subsequent targeted negation verifier, applied to the saved predictions of that version, eliminated the six false alarms of manufactured equivalence, at the cost of one correct positive classification. The results describe a promising trade-off between reducing false alarms and preserving detection. They come, however, from a set used in development and do not yet constitute evidence of generalization on new data.

Keywords: language models, rhetorical devices, false alarms, abstention, negation, contrastive evaluation.

1. The problem: identifying a device is not identifying manipulation

A system for rhetorical analysis must distinguish between the presence of a linguistic structure and its problematic use in context. A presupposition can be justified by information shared among the interlocutors. A rhetorical question can serve an explanatory role. A comparison is not, by its mere existence, deceptive.

The objective evaluated here is therefore limited: identifying the rhetorical device the author performs and endorses, according to the set's operational definitions, not establishing an intent to manipulate.

This delimitation matters for defensive tools. An incorrect label can turn an instrument of critical reading into a source of unwarranted accusations.

2. The data and the configurations analysed

The development set contains 60 contrastive pairs. Each pair includes a positive example and a control labelled as lacking the target device. The six categories, each represented by ten positive examples, are: presupposition; manufactured equivalence; false dilemma; whataboutism; loaded rhetorical question; implied causation.

Three configurations were analysed: (1) the initial classifier, used as a reference; (2) classifier v2, with explicit instructions about negation and quotation and the option to answer “insufficient_evidence”; (3) v2 with the negation check, applied exclusively to the initial manufactured-equivalence predictions. For the comparison attributable to the verifier, the v2 predictions were held fixed; the base classifier was not re-run with a different prompt.

The frozen configuration for later testing uses gpt-4o-mini, temperature 0, and is identified in the repository by the tag explanation-v2-frozen. Temperature 0 alone is not a guarantee of perfect reproducibility.

3. Evaluation indicators

The evaluation separates: false alarm (assigning a technique to a control); correct rejection (answering “none” for a control); abstention (answering “insufficient_evidence”); positive detection (assigning any technique to a positive example); correct classification (assigning the reference-label technique). Abstention is not counted as correct classification. Likewise, detecting a wrong technique is not equivalent to identifying the device correctly.

4. Results of classifier v2

Indicator	Initial classifier	V2
False alarms on controls	39/60 — 65%	16/60 — 26.7%
Controls classified correctly	21/60 — 35%	25/60 — 41.7%
Abstentions on controls	0/60	19/60 — 31.7%
Positives with technique named correctly	55/60 — 91.7%	51/60 — 85%
Detection recall	57/60 — 95%	52/60 — 86.7%

V2 reduces the false-alarm rate by 38.3 percentage points, but this reduction is accompanied by more abstentions and a drop in detection on positives. The total number of exactly-correct classifications stays the same: 76 of 120 in both configurations. The advantage of v2 therefore lies in changing the distribution of outcomes, not in increasing total exact correctness. For defensive use, replacing a wrong accusation with an abstention can have practical value.

Performance differs sharply across categories. Presupposition falls from seven to two correct classifications out of ten. By contrast, false dilemma, whataboutism and the loaded rhetorical question each keep ten correct positive classifications. On the controls, false dilemma and manufactured equivalence together account for 14 of the 16 false alarms. For the loaded rhetorical question, all ten controls produce abstentions rather than correct rejections.

5. The targeted negation check

The additional verifier examines whether the equivalence relation identified by the classifier is affirmed or denied. On the fixed v2 baseline, the reported result is: equivalence false alarms on controls 6 → 0; equivalence positives classified correctly 10 → 9; predictions outside the originally targeted class, unchanged. The introduced error appears at example Q07: the verifier treats an affirmed comparison as negated and replaces the correct label with “none”.

The result supports the local usefulness of a targeted semantic check. It does not demonstrate that the verifier understands negation in all constructions, or that the effect will hold on new texts. A traceability issue was also clarified: a run with a shorter base prompt had produced different totals; that run is not used to estimate the verifier's effect on v2.

6. Limitations and planned validation

The main limitation is the repeated use of the same 60 pairs for error analysis and system tuning. The results are descriptive for this set but may overestimate performance outside it. Other limitations are the small number of examples per category, dependence on the label definitions, and the absence of a completed independent evaluation.

The next experiment provides for a new set of 60 pairs, with different topics and constructions, labels set before running, and texts presented individually. V2 will be run unchanged, and the verifier will be applied to the saved predictions. The exploratory acceptance criteria for v2 are: at most 22 false alarms out of 60 controls; at least 48 detected out of 60 positives. For the verifier, the targets are at least a 75% relative reduction of false alarms on the negated comparisons and the loss of at most one correct classification out of ten positive equivalences; the reduction cannot be evaluated if the baseline produces no such false alarms. These thresholds are pragmatic pilot criteria, not universal standards nor sufficient tests for demonstrating generalization.

7. Conclusion

The experiment shows that correct identification of devices on positive examples can coexist with a high false-alarm rate. Introducing abstention and contextual instructions reduces these alarms but can hide significant losses in certain categories. The targeted negation check offers a favourable local result: the elimination of six false labels, at the cost of one correct positive classification. The justified conclusion is that this approach is worth testing on new data, not that the problem has been solved. For a defensive tool, performance must be judged by the balance between detection, attribution errors, and the capacity to acknowledge uncertainty.

Availability of materials

The code, label definitions, predictions, and validation protocol are public at github.com/politomorphism/politomorphism-defense. Frozen version: explanation-v2-frozen.

Addendum: Held-out evaluation performed

The frozen configuration (explanation-v2-frozen, gpt-4o-mini, temperature 0) was run unchanged on a held-out set of 60 new pairs (contrastive_pairs_v4_heldout.csv), with different topics and constructions, labelled before running. The acceptance thresholds had been recorded in the protocol before the evaluation, including for the negation verifier.

Indicator	Value	Threshold	Result
False alarms (v2)	16/60	<= 22/60	MET
Detection recall (v2)	54/60	>= 48/60	MET
Equivalence reduction (negated comparisons)	7 -> 0 (100%)	>= 75%	MET
Equivalence positives lost	0 of 10	<= 1	MET
Changes outside equivalence	0	0	MET
Total false alarms after verifier	16/60 -> 9/60	-	-

The 9 remaining false alarms: six on false dilemma (three-option controls), two on whataboutism labelled as implied causation, one on implied causation. The verifier's 7 corrections became correct rejections ('none'), not abstentions.

Verdict: The frozen configuration met all pre-set acceptance criteria on an author-built held-out set. After applying the negation verifier, total false alarms fell from 16/60 to 9/60, and recall stayed at 54/60. The result does not constitute independent validation and does not demonstrate generalization; it is a favourable result on this set, not an exclusion of random variation.

Independent validation planned

The remaining step is evaluation by independent human annotators who did not tune the system, with labels fixed beforehand and disagreements resolved. An author-built set gives an evaluation separate from the development data, but does not constitute independent validation.